

Algebra 1

Unit 13

Lesson 1 Homework

Name_____

Date_____

Period_____

The Singh family is building a pool. They were approved by their bank for a loan of \$85,000 with a simple interest rate of 2.25%.

1. Write an equation that models the Singh's loan.

2. Rewrite the equation in slope-intercept form.

3. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept		
slope		

4. What is the total the Singh's will have to pay if they pay off the loan in 5 years?

5. How many years did it take the Singh's to pay off the loan if the total they paid was \$109,000?

Sophie just graduated college and is starting her first job. She is investing money into a savings account to plan for retirement. She has \$10,000 to invest in a savings account with a 4.50% APR.

6. Write an equation that models Sophie's savings account.

7. Write the equation in slope-intercept form.

8. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept		
slope		

9. How much money will be in the savings account if Sophie wants to retire 30 years later?

10. How many years would Sophie have to work to retire with \$30,000?